

01_qc

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QC Report- Nanopore Sequencing

Sample: T1

Pre-Filtration QC

Statistics

General summary:

Mean read length:	409.8
Mean read quality:	14.6
Median read length:	330.5
Median read quality:	16.0
Number of reads:	46.0
Read length N50:	467.0
STDEV read length:	356.0
Total bases:	18,849.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	45	(97.8%)	0.0Mb
>Q15:	27	(58.7%)	0.0Mb
>Q20:	16	(34.8%)	0.0Mb
>Q25:	10	(21.7%)	0.0Mb
>Q30:	5	(10.9%)	0.0Mb

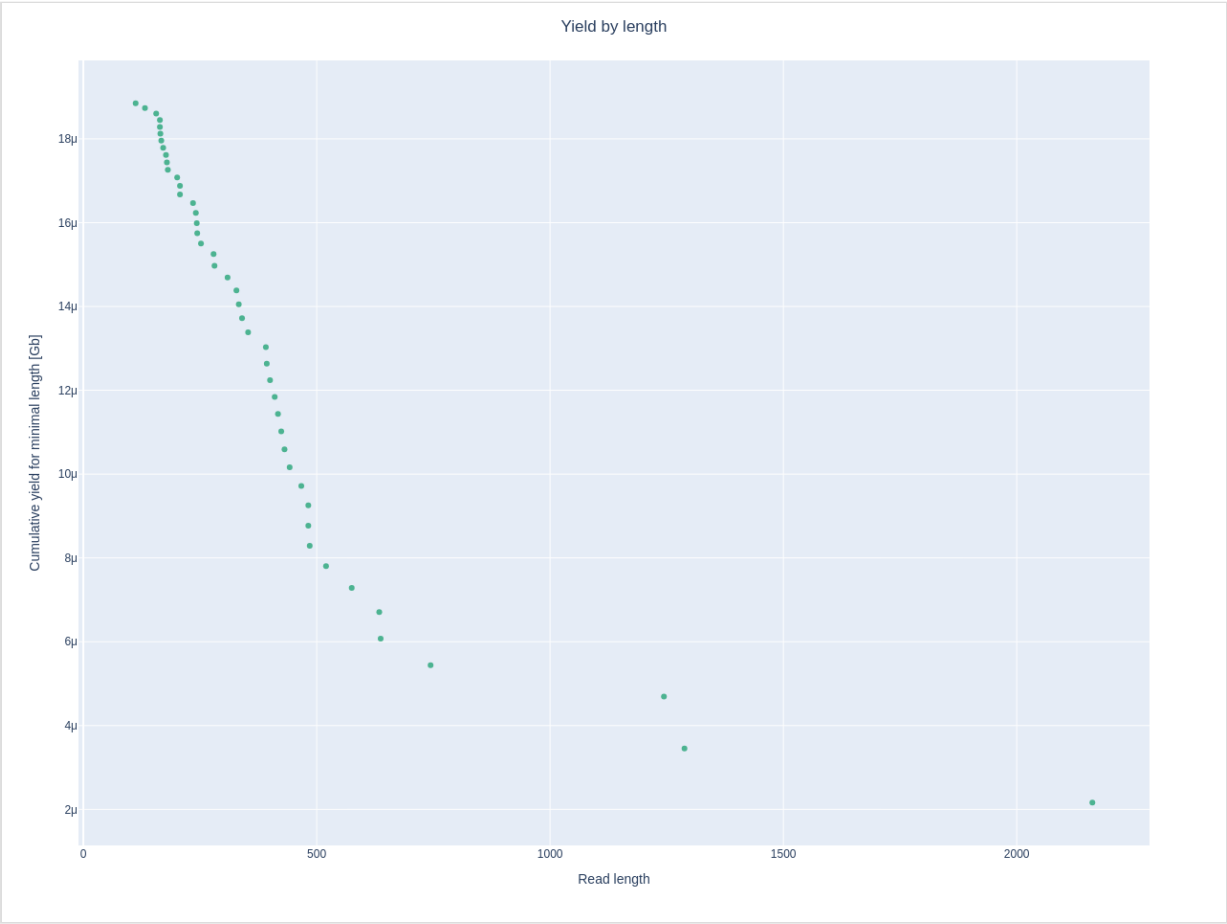
Top 5 highest mean basecall quality scores and their read lengths

1:	40.8	(132)
2:	36.2	(252)
3:	36.0	(244)
4:	32.5	(393)
5:	31.4	(243)

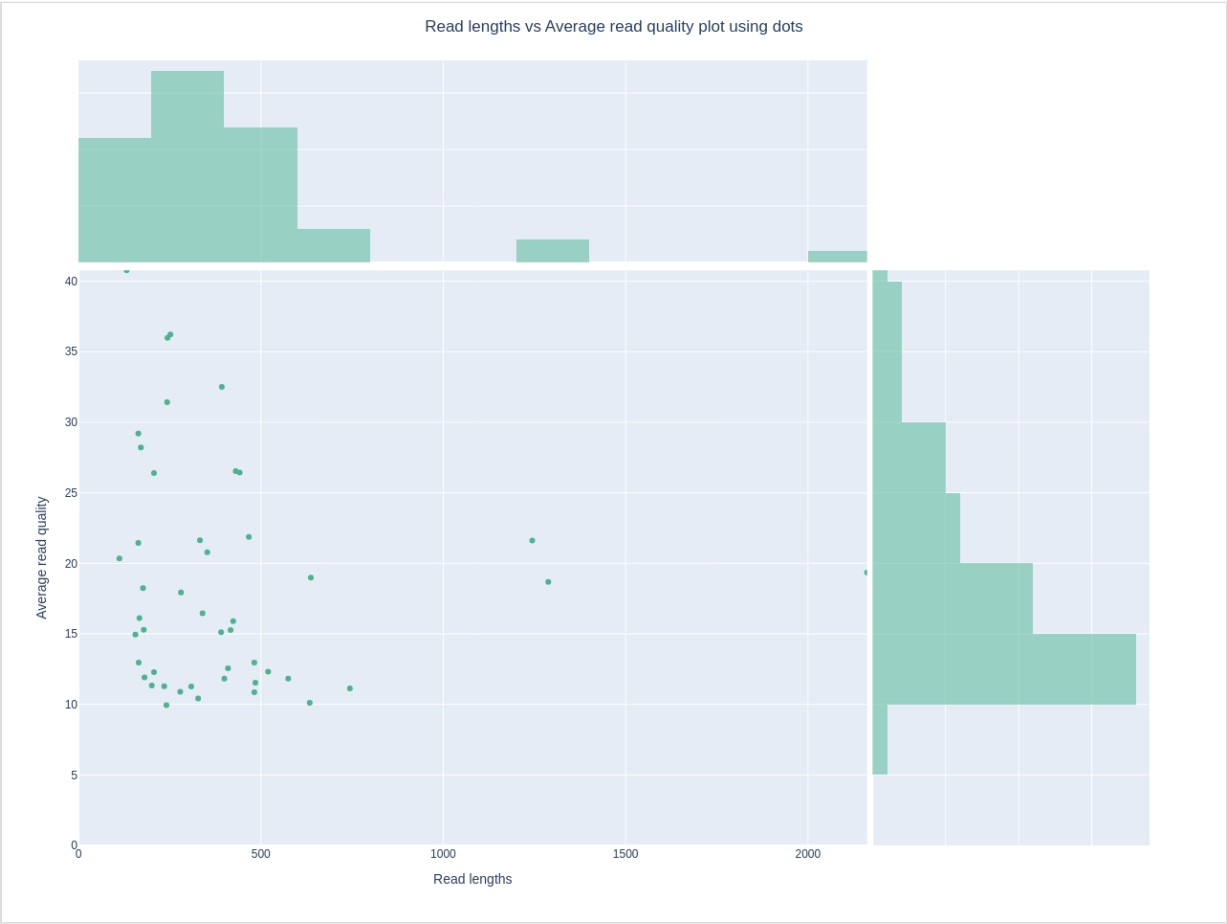
Top 5 longest reads and their mean basecall quality score

1:	2162	(19.3)
2:	1288	(18.7)
3:	1244	(21.6)
4:	744	(11.1)
5:	637	(19.0)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	975.5
Mean read quality:	13.7
Median read length:	690.5
Median read quality:	15.5
Number of reads:	8.0
Read length N50:	1,244.0
STDEV read length:	528.0
Total bases:	7,804.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	8	(100.0%)	0.0Mb
>Q15:	4	(50.0%)	0.0Mb
>Q20:	1	(12.5%)	0.0Mb
>Q25:	0	(0.0%)	0.0Mb
>Q30:	0	(0.0%)	0.0Mb

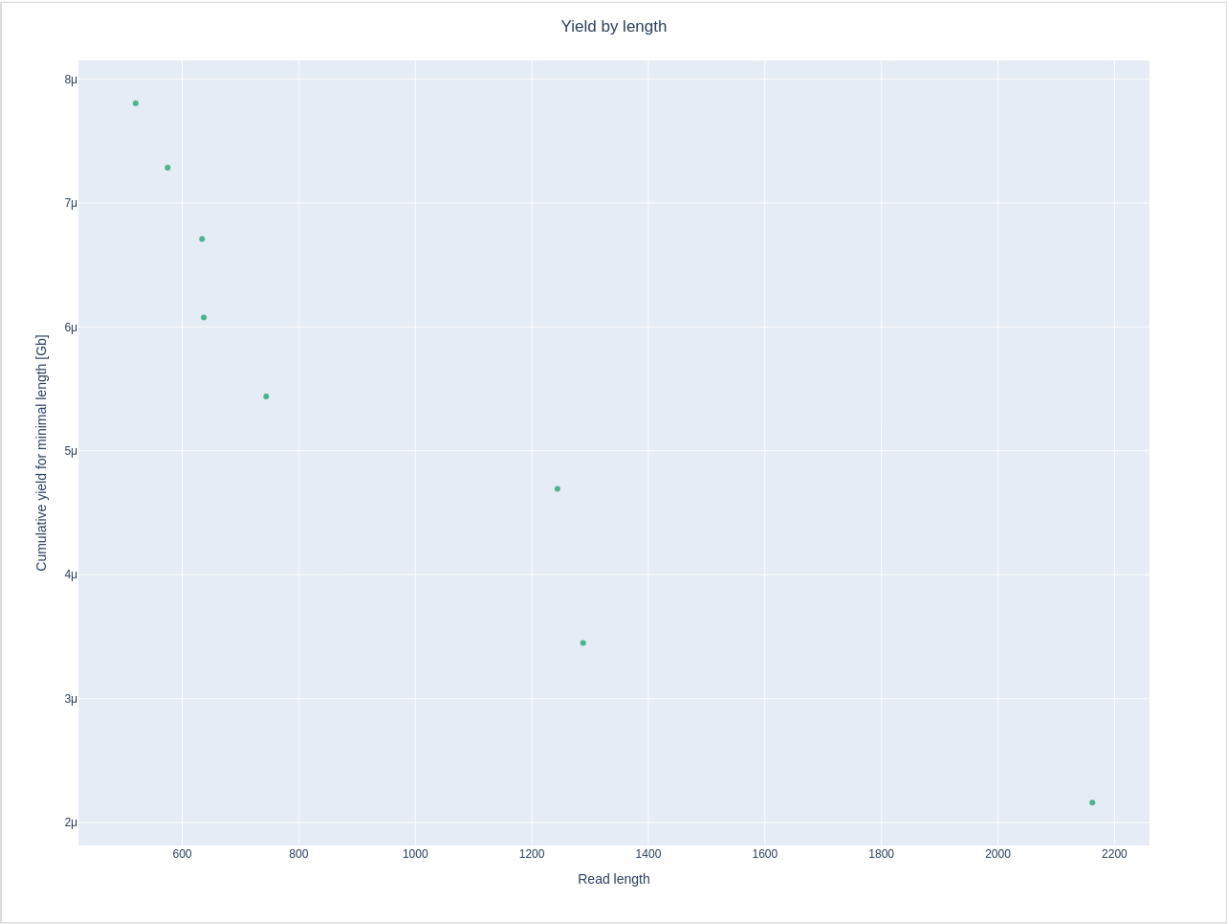
Top 5 highest mean basecall quality scores and their read lengths

1:	21.6	(1244)
2:	19.3	(2162)
3:	19.0	(637)
4:	18.7	(1288)
5:	12.3	(520)

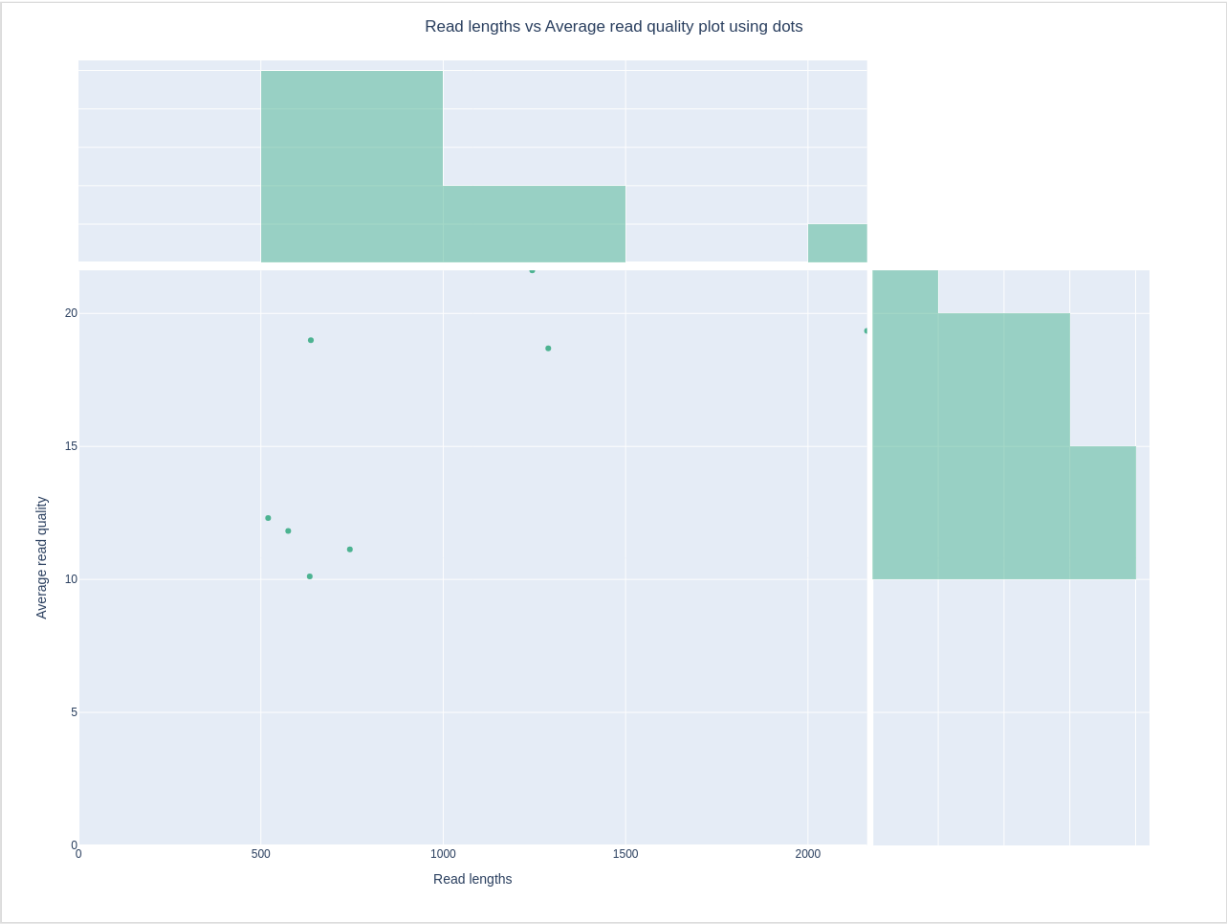
Top 5 longest reads and their mean basecall quality score

1:	2162	(19.3)
2:	1288	(18.7)
3:	1244	(21.6)
4:	744	(11.1)
5:	637	(19.0)

Yield by Length



Length vs Quality



Sample: M1

Pre-Filtration QC

Statistics

General summary:

Mean read length: 606.6

Mean read quality: 14.8

Median read length: 395.5

Median read quality: 16.4

Number of reads: 566.0

Read length N50: 749.0

STDEV read length: 966.9

Total bases: 343,348.0

Number, percentage and megabases of reads above quality cutoffs

>Q10: 550 (97.2%) 0.3Mb

>Q15: 329 (58.1%) 0.2Mb

>Q20: 176 (31.1%) 0.1Mb

>Q25: 80 (14.1%) 0.0Mb

>Q30: 27 (4.8%) 0.0Mb

Top 5 highest mean basecall quality scores and their read lengths

1: 42.1 (260)

2: 39.6 (231)

3: 39.1 (329)

4: 39.1 (200)

5: 39.1 (116)

Top 5 longest reads and their mean basecall quality score

1: 12011 (20.7)

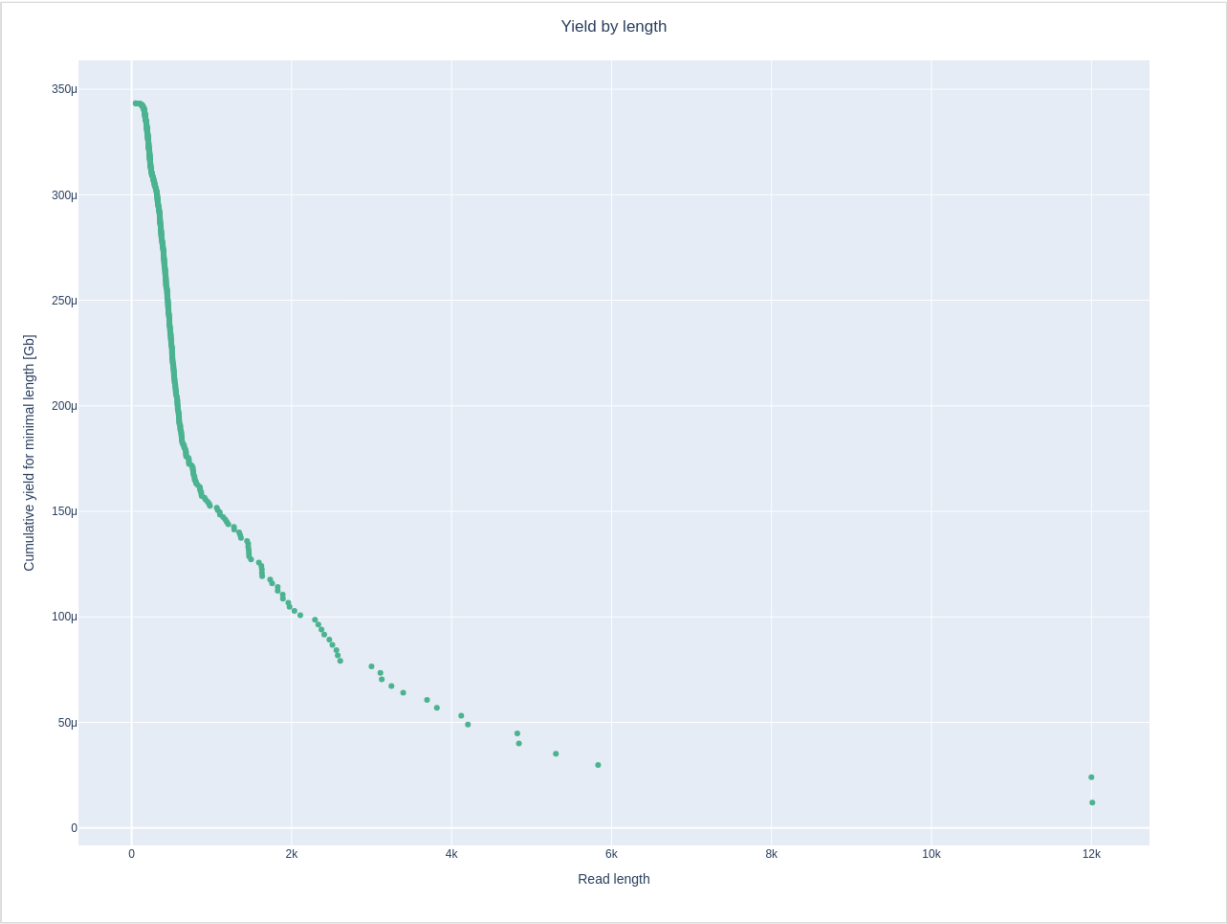
2: 11999 (18.2)

3: 5832 (12.7)

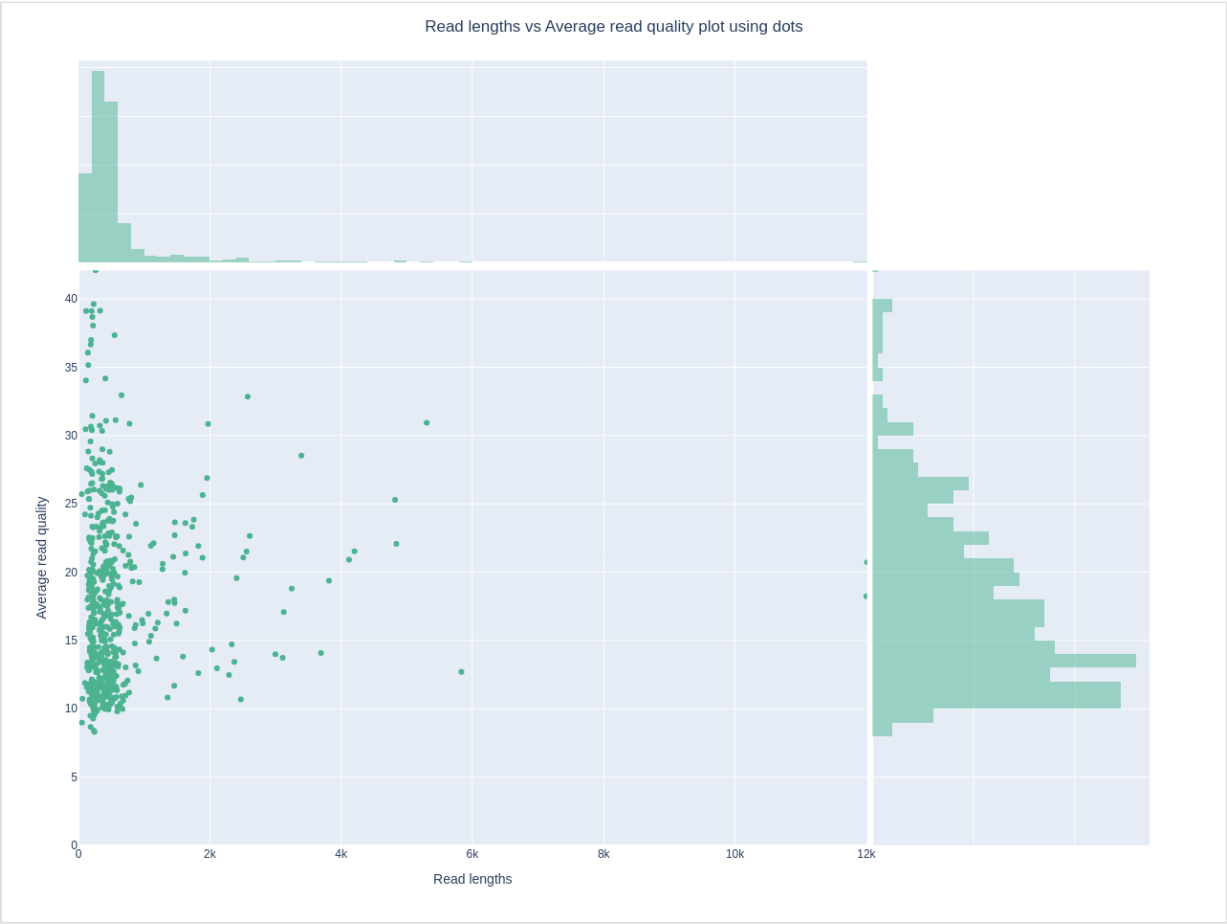
4: 5304 (30.9)

5: 4842 (22.1)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	1,261.3
Mean read quality:	15.5
Median read length:	679.0
Median read quality:	17.6
Number of reads:	181.0
Read length N50:	1,826.0
STDEV read length:	1,505.6
Total bases:	228,300.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	179 (98.9%)	0.2Mb
>Q15:	117 (64.6%)	0.2Mb
>Q20:	67 (37.0%)	0.1Mb
>Q25:	23 (12.7%)	0.0Mb
>Q30:	7 (3.9%)	0.0Mb

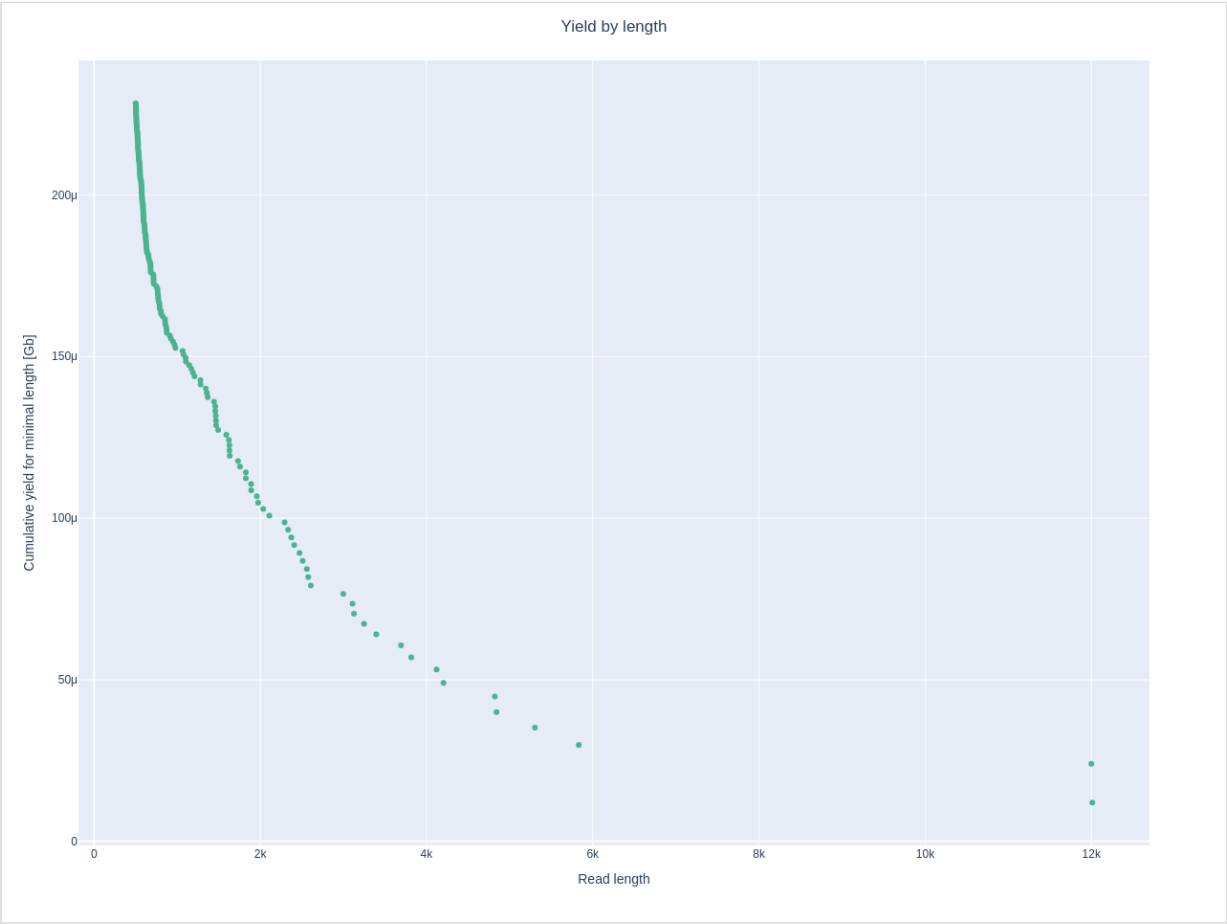
Top 5 highest mean basecall quality scores and their read lengths

1:	37.3 (550)
2:	33.0 (655)
3:	32.9 (2577)
4:	31.1 (566)
5:	30.9 (5304)

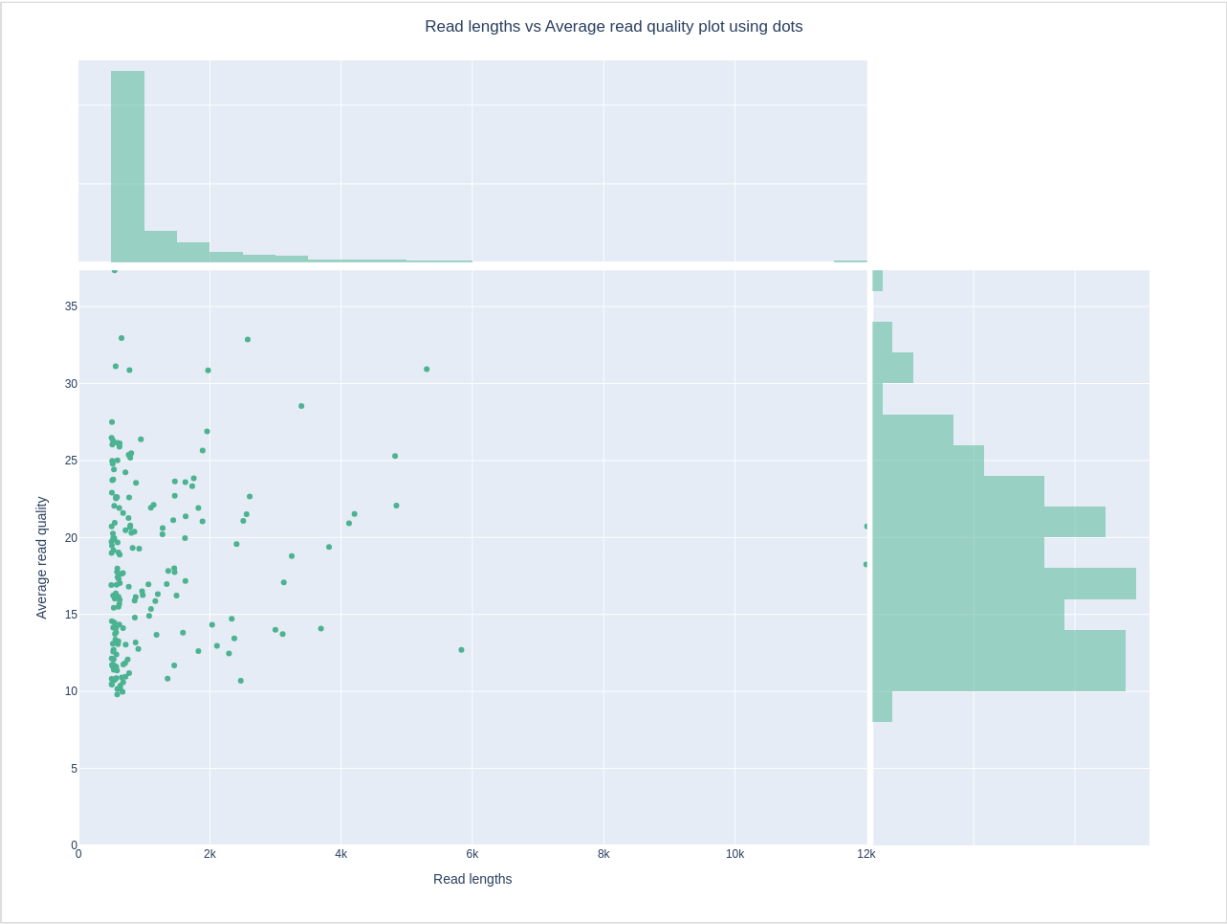
Top 5 longest reads and their mean basecall quality score

1:	12011 (20.7)
2:	11999 (18.2)
3:	5832 (12.7)
4:	5304 (30.9)
5:	4842 (22.1)

Yield by Length



Length vs Quality



Sample: A1

Pre-Filtration QC

Statistics

General summary:

Mean read length: 483.4

Mean read quality: 14.8

Median read length: 367.0

Median read quality: 16.4

Number of reads: 556.0

Read length N50: 521.0

STDEV read length: 775.8

Total bases: 268,788.0

Number, percentage and megabases of reads above quality cutoffs

>Q10: 545 (98.0%) 0.3Mb

>Q15: 326 (58.6%) 0.2Mb

>Q20: 189 (34.0%) 0.1Mb

>Q25: 80 (14.4%) 0.0Mb

>Q30: 29 (5.2%) 0.0Mb

Top 5 highest mean basecall quality scores and their read lengths

1: 42.8 (226)

2: 41.5 (202)

3: 39.6 (355)

4: 38.0 (359)

5: 37.7 (340)

Top 5 longest reads and their mean basecall quality score

1: 14540 (21.2)

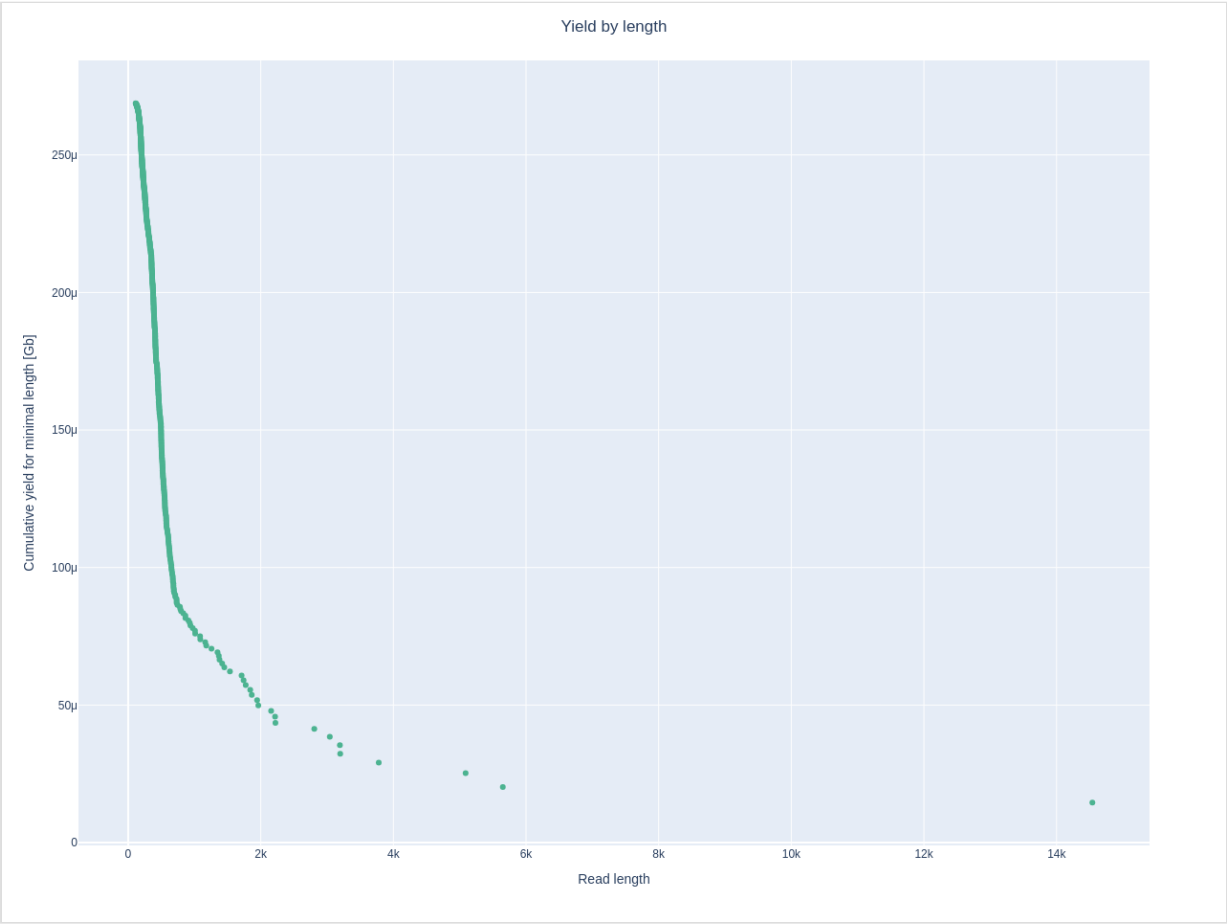
2: 5650 (10.9)

3: 5089 (23.9)

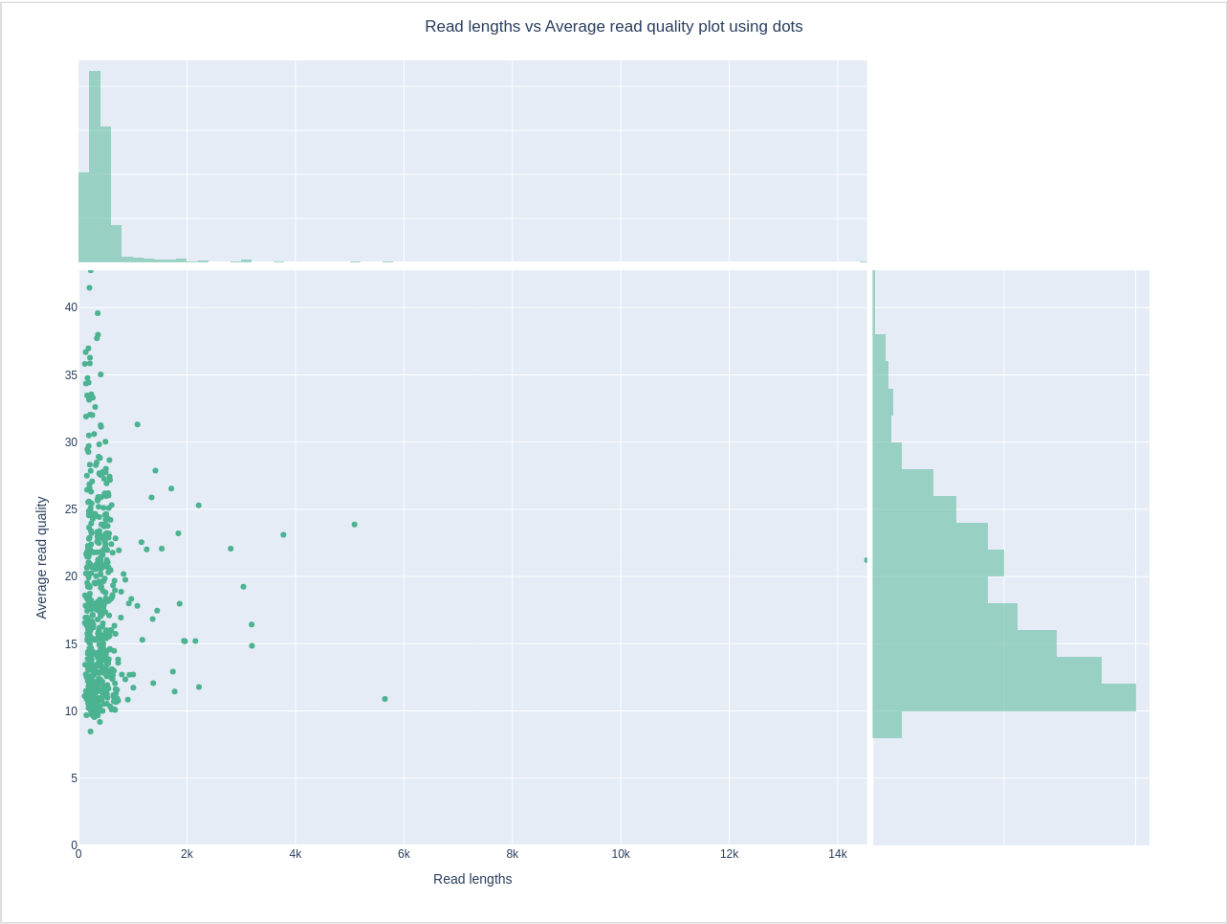
4: 3779 (23.1)

5: 3198 (14.8)

Yield by Length



Length vs Quality



Post-Filtration QC

Statistics

General summary:

Mean read length:	1,013.1
Mean read quality:	14.8
Median read length:	620.5
Median read quality:	16.0
Number of reads:	144.0
Read length N50:	1,088.0
STDEV read length:	1,382.8
Total bases:	145,893.0

Number, percentage and megabases of reads above quality cutoffs

>Q10:	144	(100.0%)	0.1Mb
>Q15:	83	(57.6%)	0.1Mb
>Q20:	48	(33.3%)	0.1Mb
>Q25:	17	(11.8%)	0.0Mb
>Q30:	1	(0.7%)	0.0Mb

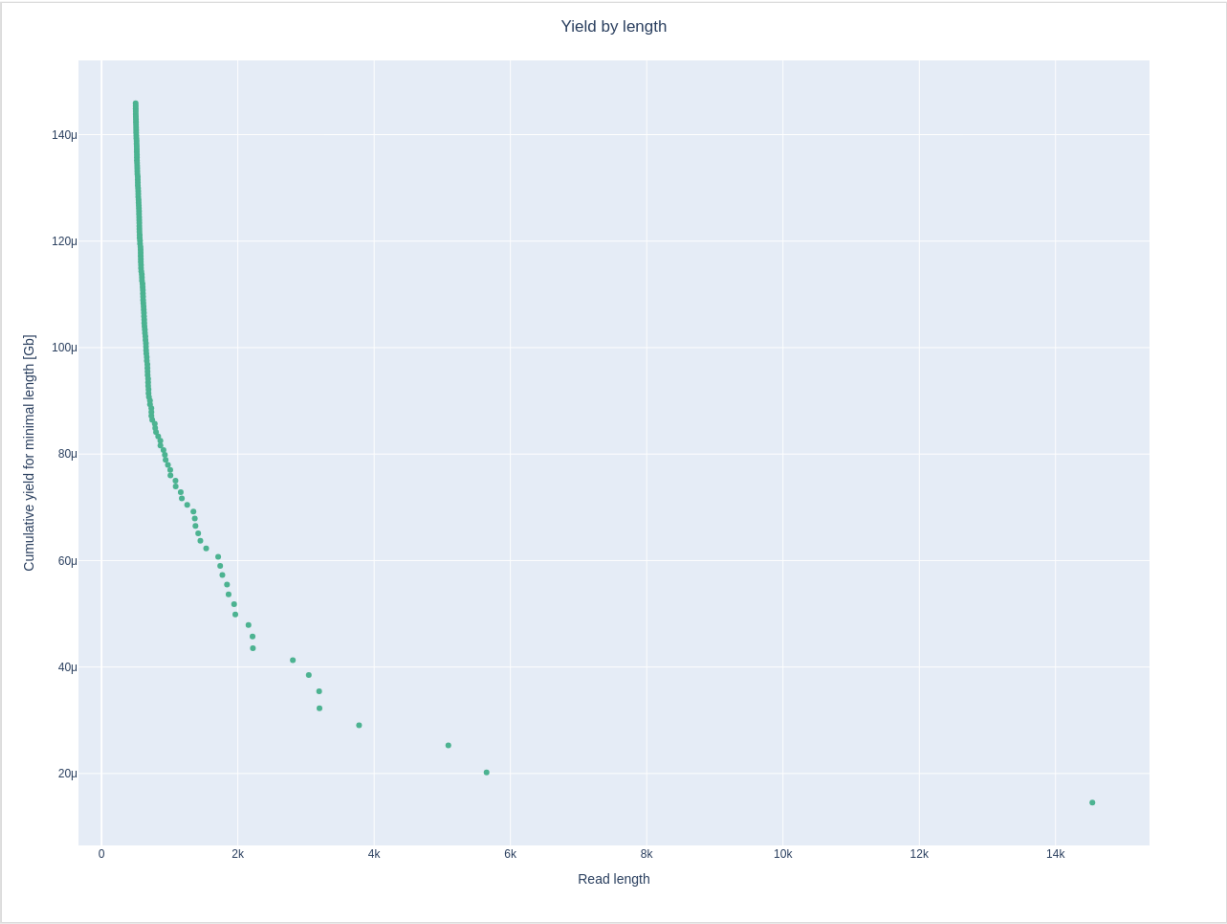
Top 5 highest mean basecall quality scores and their read lengths

1:	31.3	(1088)
2:	28.7	(571)
3:	28.0	(506)
4:	27.9	(1418)
5:	27.7	(502)

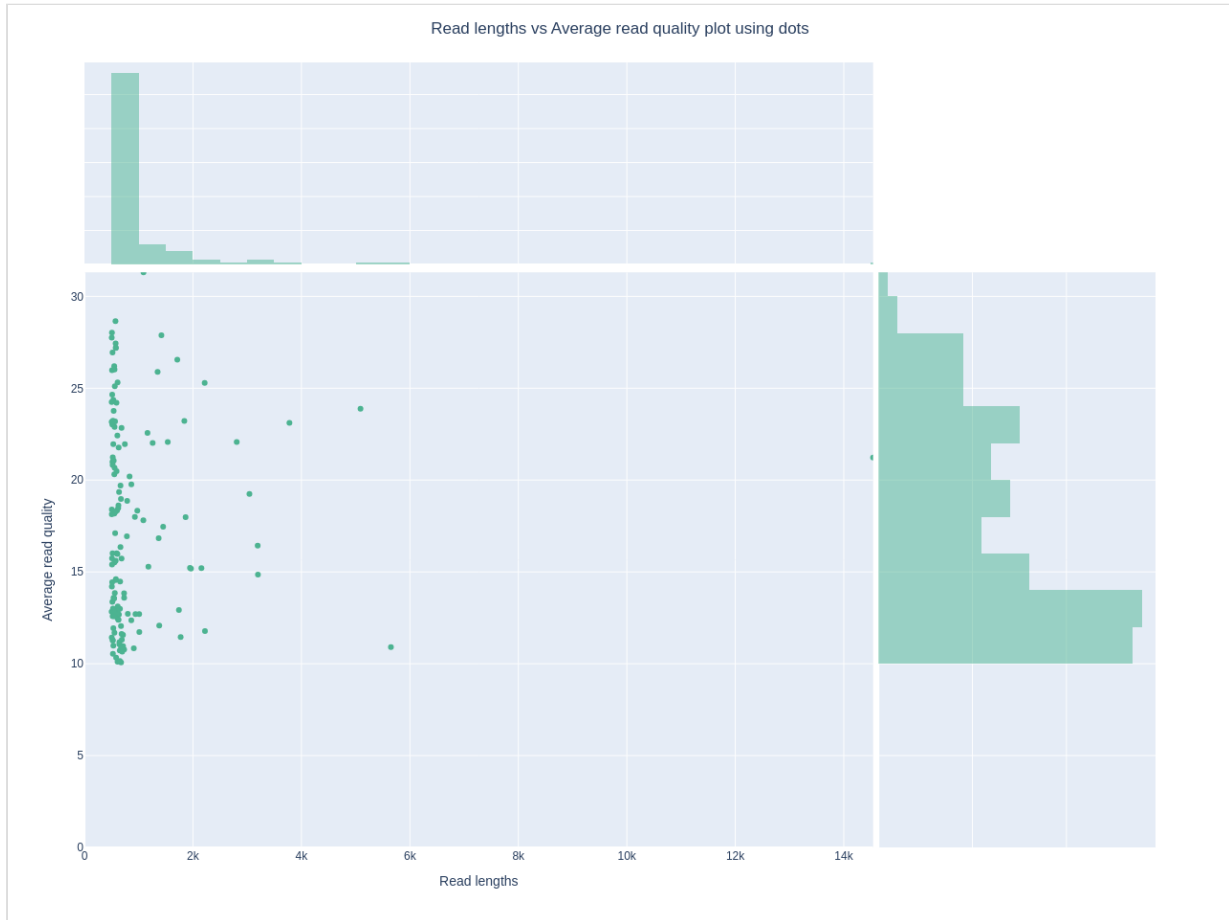
Top 5 longest reads and their mean basecall quality score

1:	14540	(21.2)
2:	5650	(10.9)
3:	5089	(23.9)
4:	3779	(23.1)
5:	3198	(14.8)

Yield by Length



Length vs Quality



Method:

QC pipeline steps:

1. Pre-QC Analysis: Raw reads analyzed with NanoPlot
2. Read Filtering: Reads filtered with FiltLong (min length: 500bp, min mean Q: 12)
3. Post-QC Analysis: Filtered reads analyzed with NanoPlot

Analysis:

Explicacion tal vez del pq de los parametros

y algun comentario adicional dependiendo que sea
bueno comentar por ejemplo

algun outlier en las muestras, los largos cortos, etc x.x.